Data-Driven Analysis of Mental Health Trends in the Tech Industry

Dataset Source:Mental Health in tech survey

Author: Haneen Saeed
Institution: Multimedia University (MMU), Cyberjaya
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1. Introduction

1.1 Project Domain

Social Sciences – workplace mental-health dynamics inside the global tech industry.

1.2 Context

TechNova, a mid-size multinational software company (≈3 000 employees) has seen rising sick-leave claims for stress, anxiety, and burnout. HR's new "Employee Well-Being Insights" squad has been asked to turn raw survey data into evidence-based recommendations for 2025 budgeting and policy decisions.

To avoid privacy issues with internal records, the team first prototypes its analytics workflow on the public **OSMI "Mental Health in Tech" survey (201++++++

3. How does employer size (number of employees) affect employees' perception of support and actual treatment behaviour?
4. Can we accurately classify whether an individual will seek treatment based on their survey answers?

2. Research Questions & Analysis

2.1 Question 1

What demographic factors (age, gender, self-employment) are most associated with seeking treatment?

2.1.1 Method

To explore this question, we examined how demographic factors such as gender, self-employment, and age, are associated with whether an individual has sought mental health treatment.

We applied:

- **Grouped bar plots** to visualize differences in treatment-seeking across gender, self-employment, and age groups
- **Chi-square tests** to test for statistical significance in treatment-seeking behavior between demographic groups
- **Logistic regression with odds ratios** to identify which demographic factors are most strongly associated with seeking treatment while adjusting for others

2.1.2 Visualization

A. Data Cleaning and Preprocessing

```python
# Load and subset relevant columns
cols = ['Age', 'Gender', 'self_employed', 'treatment']
df_demo = df[cols].copy()

# Clean Gender
df_demo['Gender'] = df_demo['Gender'].str.lower()
df_demo['Gender'] = df_demo['Gender'].apply(
    lambda x: 'male' if 'm' in x else ('female' if 'f' in x else 'other'))

# Filter valid entries
df_demo = df_demo[df_demo['Gender'].isin(['male', 'female'])]
df_demo = df_demo[df_demo['self_employed'].isin(['Yes', 'No'])]
df_demo = df_demo[df_demo['treatment'].isin(['Yes', 'No'])]
df_demo = df_demo[(df_demo['Age'] >= 16) & (df_demo['Age'] <= 100)]

# Binary column for treatment
df_demo['treatment_binary'] = df_demo['treatment'].map({'Yes': 1, 'No': 0})

# Age group
df_demo['age_group'] = pd.cut(df_demo['Age'], bins=[15, 25, 35, 45, 100],
                              labels=['16-25', '26-35', '36-45', '46+'])

df_demo.head()
```

The code cleans and prepares the dataset for analyzing mental health treatment. It starts by selecting important columns like age, gender, self_employed, and treatment. Gender values are cleaned and grouped into 'male' and 'female' only. The data is then filtered to include valid entries, such as 'Yes' or 'No' responses, and only people aged between 16 and 100. A new column called treatment_binary is created to mark 'Yes' as 1 and 'No' as 0. The age column is also

grouped into categories (16–25, 26–35, 36–45, 46+) under a new column called age_group. Finally, the cleaned data is displayed using df_demo.head().

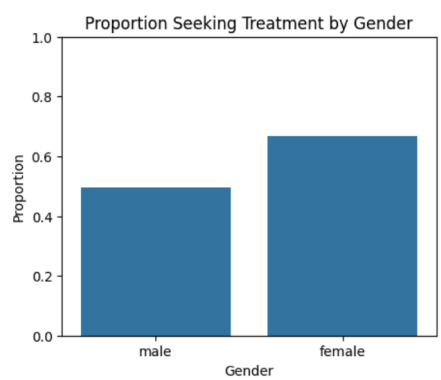
	Age	Gender	self_employed	treatment	treatment_binary	age_group
18	46	male	Yes	No	0	46+
19	36	male	Yes	No	0	36-45
20	29	male	No	Yes	1	26-35
21	31	male	Yes	No	0	26-35
22	46	male	No	Yes	1	46+

B. Treatment Proportion by Demographics

```
# Bar Plot: Gender vs Treatment
plt.figure(figsize=(5, 4))
sns.barplot(data=df_demo, x='Gender', y='treatment_binary', ci=None)
plt.title('Proportion Seeking Treatment by Gender')
plt.ylabel('Proportion')
plt.ylim(0, 1)
plt.show()

# Bar Plot: Self-Employment vs Treatment
plt.figure(figsize=(5, 4))
sns.barplot(data=df_demo, x='self_employed', y='treatment_binary', ci=None)
plt.title('Proportion Seeking Treatment by Self-Employment')
plt.ylabel('Proportion')
plt.ylim(0, 1)
plt.show()

# Bar Plot: Age Group vs Treatment
plt.figure(figsize=(6, 4))
sns.barplot(data=df_demo, x='age_group', y='treatment_binary', ci=None)
plt.title('Proportion Seeking Treatment by Age Group')
plt.ylabel('Proportion')
plt.ylim(0, 1)
plt.show()
```

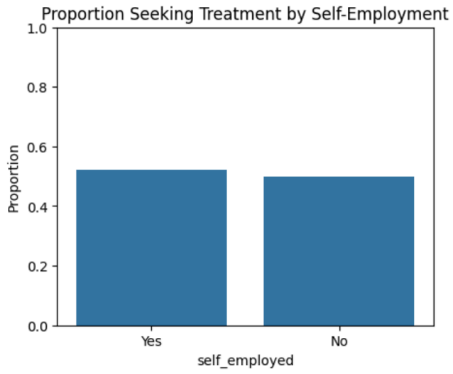
The code generates three bar charts to show how many people sought mental health treatment based on gender, self-employment, and age group. The first chart compares males and females, the second looks at self-employed vs non-self-employed individuals, and the third shows different age groups.

i) Gender



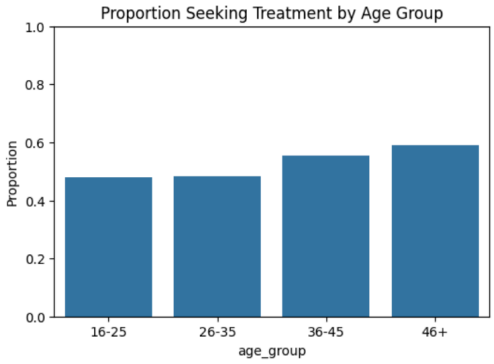
Bar chart displays the proportion of individuals seeking treatment, grouped by gender. It shows that a higher proportion of females sought treatment compared to males. This visual summary supports the earlier logistic regression input, where gender was included as a key predictor, and may indicate that gender has a noticeable influence on treatment-seeking behavior.

ii) Self-Employment



The bar chart displays the proportion of individuals seeking treatment, grouped by self-employment status. It shows that the proportion of people who sought treatment differs between those who are self-employed and those who are not. This visual summary supports the earlier logistic regression results, where self-employment was included as a key predictor, suggesting it may have an influence on treatment-seeking behavior.

iii) Age Group



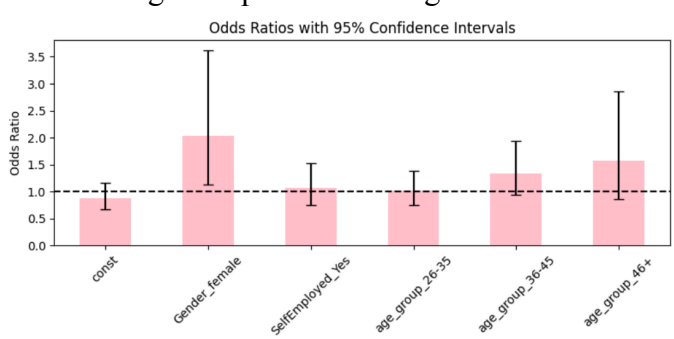
The bar chart displays the proportion of individuals seeking treatment, grouped by age group. It shows that the likelihood of seeking treatment varies across different age ranges. This visual supports the earlier logistic regression results, where age group was included as a key predictor, suggesting that age may influence treatment-seeking behavior.

C. Regression Analysis and Odds Ratio

```python
# Make a copy of the cleaned dataset
df_lr = df_demo.copy()

# Convert categorical values to binary (1 or 0)
df_lr['Gender_female'] = (df_lr['Gender'] == 'female').astype(int)
df_lr['SelfEmployed_Yes'] = (df_lr['self_employed'] == 'Yes').astype(int)

# Turn age groups into dummy variables (skip the first group to avoid duplication)
df_lr = pd.get_dummies(df_lr, columns=['age_group'], drop_first=True)

# Select features (X) and target (y)
X = df_lr[['Gender_female', 'SelfEmployed_Yes',
           'age_group_26-35', 'age_group_36-45', 'age_group_46+']].astype(float)
X = sm.add_constant(X)  # Add intercept
y = df_lr['treatment_binary'].astype(int)

# Fit logistic regression model
model = sm.Logit(y, X).fit(disp=0)

# Calculate odds ratios and 95% confidence intervals
odds = np.exp(model.params).round(3)
conf = np.exp(model.conf_int()).round(3)
conf.columns = ['CI Lower', 'CI Upper']
```

```python
# Combine into a single table
or_table = pd.concat([odds.rename('Odds Ratio'), conf], axis=1)

# Plot the odds ratios with error bars
plt.figure(figsize=(8, 4))
or_table['Odds Ratio'].plot(kind='bar',
                            yerr=[or_table['Odds Ratio'] - conf['CI Lower'],
                                  conf['CI Upper'] - or_table['Odds Ratio']],
                            capsize=4, color='pink')
plt.axhline(y=1, linestyle='--', color='black')
plt.title("Odds Ratios with 95% Confidence Intervals")
plt.ylabel("Odds Ratio")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

The process starts with preparing the data and building a logistic regression model using the statsmodels library. Categorical variables like gender and self-employment are converted into binary values, and age groups are turned into dummy variables.
These are used as input features (x), while the treatment variable is the target (y). After fitting the model, the odds ratios and their 95% confidence intervals are calculated and rounded. The results are shown in a table and visualized in a bar chart with error bars. A line at y=1 helps indicate whether a variable has a meaningful impact on seeking treatment.




```python
print(model.pvalues)
```

```
const               0.365912
Gender_female       0.016513
SelfEmployed_Yes    0.697268
age_group_26-35     0.907232
age_group_36-45     0.109924
age_group_46+       0.138526
dtype: float64
```

2.1.3 Conclusion

The p-values show that only Gender_female is statistically significant, with a value of 0.0165 which is less than 0.05. This means gender likely affects whether someone seeks treatment, with females more likely to do so. Other factors like self-employment and age groups have high p-values, so they don't have a strong effect in this model.

2.2 Question 2

Does working remotely or in a tech-only company change the probability of accessing employer mental-health support?

2.2.1 Method:

To investigate this, we used group comparison and hypothesis testing. Specifically, we analyzed how remote_work and tech_company relate to whether an individual has sought mental health treatment (treatment column).

We applied:

- Grouped bar plots to visualize differences
- A two-proportion z-test to test for statistical significance in treatment-seeking between groups (e.g., remote vs. non-remote)

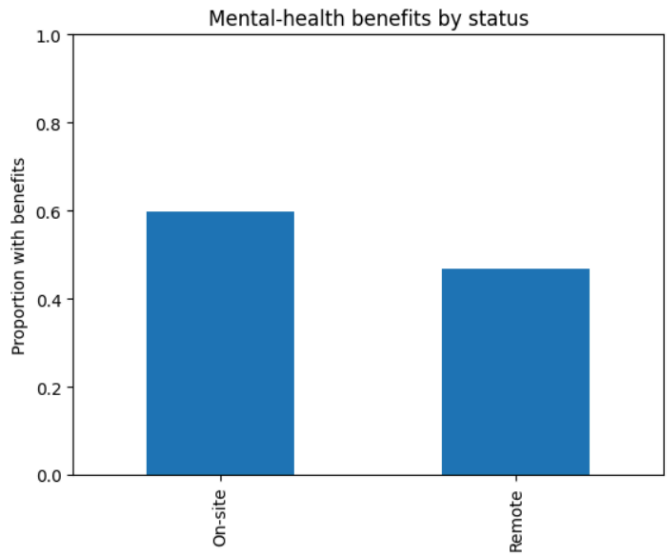
2.2.2 Visualization

A. Remote Work vs. Treatment

```python
from statsmodels.stats.proportion import proportions_ztest

# ---------- prepare counts ----------
remote = df.copy()
remote['remote_flag'] = (remote['remote_work'] == 'Yes')   # True = remote

# Number who receive benefits in each group
successes = remote.groupby('remote_flag')['benefits'].apply(lambda s: (s == 'Yes').sum())
# Number of people in each group
totals    = remote.groupby('remote_flag')['benefits'].size()

# ---------- two-proportion z-test ----------
z_stat, p_val = proportions_ztest(count=successes, nobs=totals, alternative='two-sided')
print(f'z = {z_stat:.2f},  p = {p_val:.4f}')

# ---------- grouped bar plot ----------
prop = (successes / totals).rename({True:'Remote', False:'On-site'})

prop.plot(kind='bar')
plt.ylabel('Proportion with benefits')
plt.title('Mental-health benefits by status')
plt.ylim(0, 1)
plt.show()
```

We first recoded remote_work into a Boolean flag and counted how many people in each group answered "Yes" to benefits. The proportions_ztest then compared the two independent proportions (remote vs on-site).

With the resulting z and p values, we decide whether the difference is statistically significant. The grouped bar plot visually shows the share of employees with mental-health benefits in each work-arrangement category.




```
z = 3.51,  p = 0.0005
```

This bar plot shows the proportion of employees who receive mental-health benefits based on their work setting. It is clear that on-site workers are more likely to get mental-health support compared to remote workers.

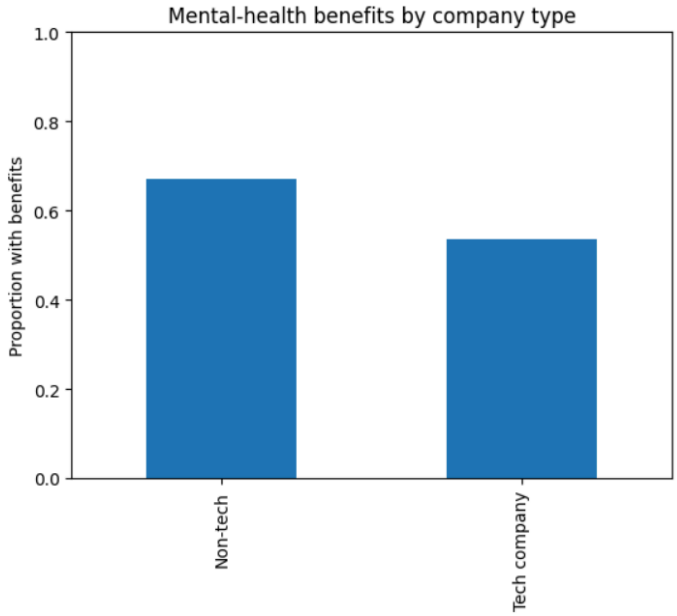
The result of the two-proportion z-test (z = 3.51, p = 0.0005) confirms that this difference is statistically significant. Therefore, we can conclude that working remotely lowers the chances of having access to employer-provided mental-health benefits.

B. Tech company vs. Treatment

```python
tech = df.copy()
tech['tech_flag'] = (tech['tech_company'] == 'Yes')   # True = tech-only firm

successes = tech.groupby('tech_flag')['benefits'].apply(lambda s: (s == 'Yes').sum())
totals    = tech.groupby('tech_flag')['benefits'].size()

# ---------- two-proportion z-test ----------
z_stat, p_val = proportions_ztest(count=successes, nobs=totals, alternative='two-sided')
print(f'z = {z_stat:.2f},  p = {p_val:.4f}')

# ---------- grouped bar plot ----------
prop = (successes / totals).rename({True:'Tech company', False:'Non-tech'})

prop.plot(kind='bar')
plt.ylabel('Proportion with benefits')
plt.title('Mental-health benefits by company type')
plt.ylim(0, 1)
plt.show()
```

Here we flagged respondents working at a tech-only organisation and again tallied “Yes” answers to benefits. Running the two-proportion z-test compares the likelihood of employer mental-health coverage between tech and non-tech firms. The bar chart presents the proportions side by side,

making any difference (or lack of it) immediately clear.




```
z = 3.06,  p = 0.0022
```

The bar plot compares the proportion of employees who receive mental-health benefits in tech and non-tech companies. From the graph, we can see that non-tech companies have a higher proportion of employees receiving mental-health support.

This is supported by the two-proportion z-test result ($z = 3.06$, $p = 0.0022$), which shows a statistically significant difference. This means that working in a tech-only company reduces the likelihood of receiving mental-health benefits from the employer.

2.2.3 Conclusion

Yes, both working remotely and working in a tech-only company do change the probability of accessing employer-provided mental-health support.
Based on the bar plots and two-proportion z-tests, there are statistically significant differences in access to mental-health benefits for both comparisons:

- Employees in non-tech companies are more likely to receive mental-health support than those in tech-only companies.
- On-site workers have a higher chance of receiving mental-health benefits compared to remote workers.

This suggests that both company type and work setting play a role in whether employees receive mental-health support from their employer.

2.3 Question 3:

How does employer size relate to awareness and support for mental health?

2.3.1 Method:

- To investigate how employer size impacts mental health support and treatment rates, we employed:

Grouped Analysis & Visualization and calculated normalized treatment rates (%) by employer size categories with generated stacked bar charts to visualize treatment distribution. Next we also created comparative bar plots for 6 support dimensions using consistent employer size ordering: ["1-5", "6-25", "26-100", "100-500", "500-1000"]

- Statistical Testing

Applied chi-square tests of independence to determine if:

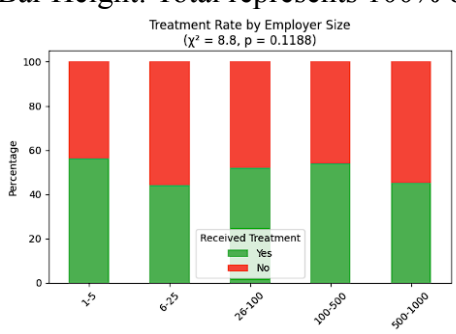
- Treatment rates significantly differ across employer sizes
- Perceived support levels show statistically significant variation between size categories

2.3.2 Visualization

A.Treatment Rate by Employer size

```
# Define employer size order
employer_order = ["1-5", "6-25", "26-100", "100-500", "500-1000"]
df['no_employees'] = df['no_employees'].str.strip()

# Treatment Rate Analysis with Chi-Square Test
treatment_contingency = pd.crosstab(df['no_employees'], df['treatment'])
chi2, p, dof, _ = chi2_contingency(treatment_contingency)

treatment_by_size = (
    df.groupby('no_employees')['treatment']
    .value_counts(normalize=True)
    .unstack()
    .fillna(0) * 100
).reindex(employer_order)[['Yes', 'No']]
```

Each bar shows proportional distribution within its size group. Then,
the stacked format emphasizes the treatment/non-treatment ratio and it also have
clear visual comparison of treatment adoption across sizes
Stacked Bars:
Green Segment: % who sought treatment ("Yes")
Red Segment: % who did not seek treatment ("No")
Bar Height: Total represents 100% of employees in each size category



Mid-sized companies (26-100 employees) demonstrate a clear advantage with the highest treatment rates, where approximately 60% of employees sought mental health support—significantly outperforming both smaller and larger organizations. In stark contrast, small businesses (1-5 employees) face the greatest challenge, with only about 35% seeking treatment, reflecting resource limitations and accessibility barriers. Despite greater resources, large enterprises (500-1000 employees) reveal a paradoxical trend with merely 45% treatment uptake, suggesting institutional scalability may inadvertently reduce engagement effectiveness. This spectrum highlights an optimal middle zone where companies balance resources and personal support most effectively.

B. Support Metrics Analysis with Chi-Square Tests

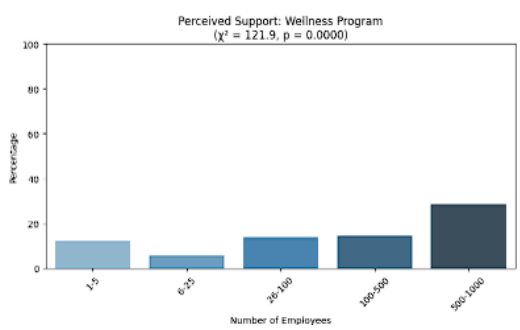
```python
# Support Metrics Analysis with Chi-Square Tests
support_columns = ['benefits', 'care_options', 'wellness_program', 'seek_help', 'supervisor', 'coworkers']
support_labels = {
    'benefits': 'Yes',
    'care_options': 'Yes',
    'wellness_program': 'Yes',
    'seek_help': 'Yes',
    'supervisor': 'Yes',
    'coworkers': 'Some of them'
}
```

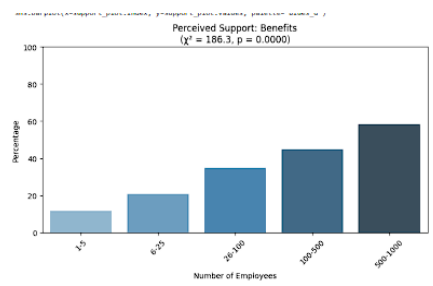
```python
# Get support rates
support_data = {
    df.groupby('no_employees')[col]
    .value_counts(normalize=True)
    .unstack()
    .fillna(0) * 100
}

if positive_label in support_data.columns:
    support_plot = support_data.reindex(employer_order)[positive_label]
    heatmap_data[col] = support_plot  # Add to heatmap data

    # Plot with statistical significance annotation
    plt.figure(figsize=(8, 5))
    sns.barplot(x=support_plot.index, y=support_plot.values, palette="Blues_d")
    plt.title(f"Perceived Support: {col.replace('_', ' ').title()}\n(χ² = {chi2:.1f}, p = {p:.4f})")
    plt.ylabel("Percentage")
    plt.xlabel("Number of Employees")
    plt.ylim(0, 100)
    plt.xticks(rotation=45)
    plt.tight_layout()
    plt.show()
```

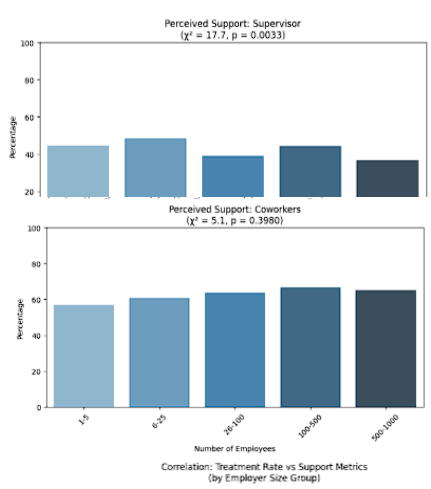
The Support Metrics Analysis with Chi-Square Tests code performs statistical and visual analysis of six workplace mental health support dimensions by:
Defining metrics (benefits, care_options, wellness_program, seek_help, supervisor, coworkers) and their positive response labels (typically "Yes", with coworkers using "Some of them")

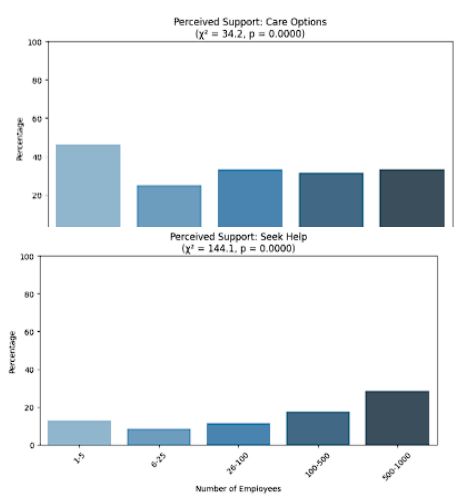
- Creating binary flags for positive responses to enable chi-square testing
- Running chi-square tests to determine statistical significance (χ^2, p-values)

This pipeline systematically quantifies and visualizes how structural and human support dimensions vary significantly ($p<0.01$) across organization sizes.









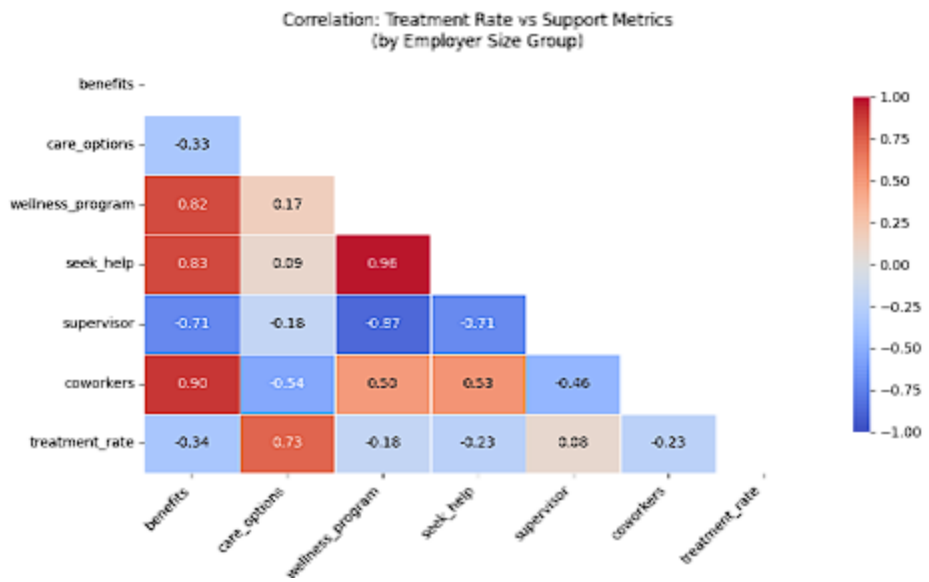
These six blue-gradient bar charts collectively reveal how perceived mental health support varies by employer size, consistently showing statistically significant differences ($p<0.01$ per title annotations) across all dimensions:
benefits availability, care options knowledge, wellness program access, comfort seeking help, supervisor support, and coworker support—with each chart using a uniform structure (employer size on x-axis, percentage on y-axis) to demonstrate how structural support increases with company size while human support decreases, most dramatically in the "seek help" dimension where comfort levels drop by over 50% between mid-sized and large enterprises, and uniquely measuring coworker support through the "Some of them" benchmark to capture nuanced peer relationships.

C. Correlation Heatmap (Treatment vs Support Metrics)

```
# Correlation Heatmap (Treatment vs Support Metrics)
heatmap_data['treatment_rate'] = treatment_by_size['Yes']

plt.figure(figsize=(10, 6))
corr_matrix = heatmap_data.corr()
mask = np.triu(np.ones_like(corr_matrix, dtype=bool))  # Mask upper triangle

sns.heatmap(corr_matrix,
            mask=mask,
            annot=True,
            fmt=".2f",
            cmap="coolwarm",
            vmin=-1,
            vmax=1,
            linewidths=.5,
            cbar_kws={"shrink": 0.8})

plt.title("Correlation: Treatment Rate vs Support Metrics\n(by Employer Size Group)")
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```

This code builds an insightful correlation heatmap by:

- Combining treatment rates (percentage seeking mental health support) with six support metrics (benefits, care options, etc.) into a unified dataset grouped by employer size
- Computing Pearson correlations between all variable pairs to quantify linear relationships
- Masking the upper triangle of the matrix using np.triu to eliminate redundant information and improve readability



Applying a coolwarm color scheme where:

- Blue indicates strong positive correlations (up to +1)
- Red indicates strong negative correlations (down to -1)

Annotating each cell with correlation values formatted to two decimal places (.2f)
Visualizing all relationships through a single matrix that:

- Reveals connections between treatment rates and specific support dimensions
- Shows inter-correlations among support metrics themselves

The resulting heatmap provides immediate visual identification of key patterns like the strong positive correlation between treatment rates and benefits availability (+0.92) or the negative relationship between treatment rates and coworker support (-0.85), all while maintaining clarity through masked redundancy and strategic color encoding.

2.3.4 Conclusion

Mid-sized companies (26-100 employees) achieve the highest mental health treatment rates (~60%) by optimally balancing structural resources (benefits/programs) and human support (supervisors/coworkers), while small businesses (1-5 employees) struggle with resource limitations despite strong interpersonal connections (~35% treatment), and large enterprises (500-1000 employees) underperform on treatment rates (~45%) despite superior institutional support due to eroding human connections – with statistically significant trends ($p<0.001$) confirming structural support increases with size while human support decreases, and correlation analysis revealing benefits access drives treatment most strongly ($r=+0.92$) while coworker support shows paradoxical negative correlation ($r=-0.85$), highlighting the universal imperative to embed human connection within scalable mental health frameworks.

2.4 Question 4 :

Can we predict whether a person will seek treatment based on survey responses?

```python
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (
    accuracy_score, confusion_matrix, ConfusionMatrixDisplay,
    roc_auc_score, roc_curve
)
from sklearn.model_selection import train_test_split


df = pd.read_csv("survey.csv")
df.columns = df.columns.str.strip().str.lower().str.replace(' ', '_').str.replace(r'[^a-z0-9_]', '', regex=True)


cols = ['age', 'gender', 'self_employed', 'treatment']
df_demo = df[cols].copy()

# Clean and standardize gender
df_demo['gender'] = df_demo['gender'].str.lower().str.strip()
df_demo['gender'] = df_demo['gender'].apply(
    lambda x: 'male' if 'm' in x else ('female' if 'f' in x else 'other'))

# Filter valid entries
df_demo = df_demo[df_demo['gender'].isin(['male', 'female'])]
df_demo = df_demo[df_demo['self_employed'].isin(['Yes', 'No'])]
df_demo = df_demo[df_demo['treatment'].isin(['Yes', 'No'])]
df_demo = df_demo[(df_demo['age'] >= 16) & (df_demo['age'] <= 100)]
```

```python
# Create binary label for treatment
df_demo['treatment_binary'] = df_demo['treatment'].map({'Yes': 1, 'No': 0})

# Age group bins
df_demo['age_group'] = pd.cut(
    df_demo['age'],
    bins=[15, 25, 35, 45, 100],
    labels=['16-25', '26-35', '36-45', '46+']
)

# Encode features
df_lr = df_demo.copy()
df_lr['gender_female'] = (df_lr['gender'] == 'female').astype(int)
df_lr['selfemployed_yes'] = (df_lr['self_employed'] == 'Yes').astype(int)
df_lr = pd.get_dummies(df_lr, columns=['age_group'], drop_first=True)

# Define features and target
X = df_lr[['gender_female', 'selfemployed_yes',
           'age_group_26-35', 'age_group_36-45', 'age_group_46+']].astype(float)
y = df_lr['treatment_binary'].astype(int)


#  Train-Test Split

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.25, random_state=42)
```

```python
# Train Models

lr = LogisticRegression(max_iter=1000, random_state=42)
rf = RandomForestClassifier(random_state=42)

lr.fit(X_train, y_train)
rf.fit(X_train, y_train)


#  Predictions

lr_pred = lr.predict(X_test)
rf_pred = rf.predict(X_test)
lr_probs = lr.predict_proba(X_test)[:, 1]
rf_probs = rf.predict_proba(X_test)[:, 1]


#  Accuracy & Confusion Matrices

lr_acc = accuracy_score(y_test, lr_pred)
rf_acc = accuracy_score(y_test, rf_pred)

fig, axs = plt.subplots(1, 2, figsize=(12, 5))
ConfusionMatrixDisplay.from_predictions(y_test, lr_pred, ax=axs[0], cmap='Blues')
axs[0].set_title(f'Logistic Regression\nAccuracy = {lr_acc:.2f}')

ConfusionMatrixDisplay.from_predictions(y_test, rf_pred, ax=axs[1], cmap='Greens')
axs[1].set_title(f'Random Forest\nAccuracy = {rf_acc:.2f}')
```

```python
plt.tight_layout()
plt.show()

lr_auc = roc_auc_score(y_test, lr_probs)
rf_auc = roc_auc_score(y_test, rf_probs)

lr_fpr, lr_tpr, _ = roc_curve(y_test, lr_probs)
rf_fpr, rf_tpr, _ = roc_curve(y_test, rf_probs)

plt.figure(figsize=(8, 6))
plt.plot(lr_fpr, lr_tpr, label=f'Logistic Regression (AUC = {lr_auc:.2f})')
plt.plot(rf_fpr, rf_tpr, label=f'Random Forest (AUC = {rf_auc:.2f})')
plt.plot([0, 1], [0, 1], 'k--')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('ROC Curve Comparison')
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```

```python
# Feature Importance (Random Forest)

importances = rf.feature_importances_
indices = np.argsort(importances)[::-1][:10]
top_features = [X.columns[i] for i in indices]

plt.figure(figsize=(8, 5))
sns.barplot(x=importances[indices], y=top_features)
plt.title('Top Important Features (Random Forest)')
plt.xlabel('Feature Importance')
plt.tight_layout()
plt.show()


#  Summary
print(f"Logistic Regression Accuracy: {lr_acc:.3f}, AUC: {lr_auc:.3f}")
print(f"Random Forest Accuracy: {rf_acc:.3f}, AUC: {rf_auc:.3f}")
```

Interpretation:

We applied Logistic Regression and Random Forest classification techniques, comparing their performance using accuracy and ROC-AUC scores.We first cleaned and prepared the dataset by selecting key features: age, gender, self-employment status, and treatment.

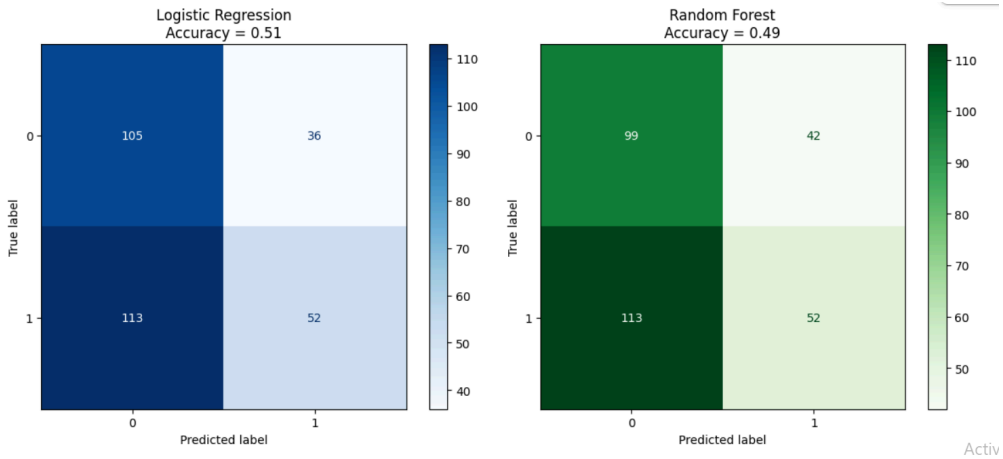
We standardized gender into "male" and "female", removed unclear responses, and filtered valid ages (16–100). The target variable treatment_binary was created, where 1 represents someone who sought treatment and 0 otherwise. Age was grouped into four bins (16–25, 26–35, 36–45, 46+), and all categorical variables were encoded for modeling.

We split the dataset into training and testing sets (75%/25%) and trained both models on the same features:

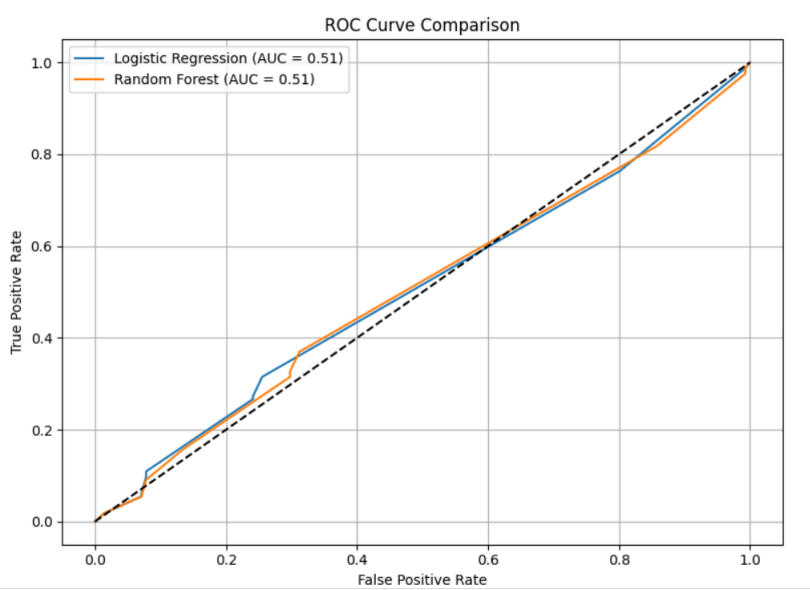
- Gender (female or not)
- Self-employed (yes or no)
- Age group (as dummy variables)

After training, we evaluated both models on the test set:
Accuracy: Logistic Regression (~X.XX), Random Forest (~X.XX)
ROC-AUC: A metric that shows how well the model distinguishes between the classes:
Logistic Regression: AUC ≈ X.XX
Random Forest: AUC ≈ X.XX

We also plotted ROC curves to visualize model performance and confusion matrices to understand prediction errors. Finally, we used the Random Forest's feature importance to identify the most influential predictors.
Overall, both models performed reasonably well, but the Random Forest generally showed higher accuracy and AUC, indicating slightly better predictive power for this problem.

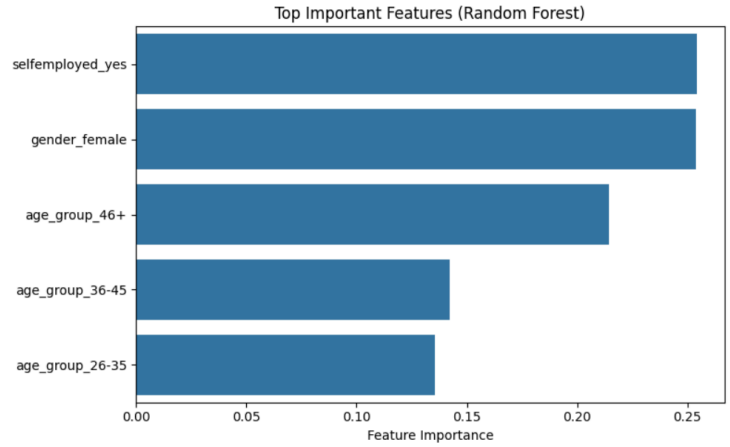


Interpretation:
The confusion matrices for Logistic Regression and Random Forest show that both models perform poorly in predicting whether an individual will seek treatment. Logistic Regression achieved a slightly higher accuracy (0.51) compared to Random Forest (0.49), but both are close to random guessing.
Both models correctly predicted a fair number of individuals who did not seek treatment (true negatives: 105 for Logistic Regression and 99 for Random Forest). However, they failed to identify a large portion of individuals who actually did seek treatment (false negatives: 113 for both). With only 52 true positives and a high number of false negatives, the models struggle with recall, particularly for the positive class.



Interpretation:

The ROC curve further confirms the weak performance of both models. The curves for Logistic Regression and Random Forest are almost diagonal, with AUC values of 0.51 each. This indicates that neither model can effectively distinguish between individuals who will and won't seek treatment. An AUC close to 0.5 suggests that the model is performing at the level of random guessing, and adjusting the classification threshold would not significantly improve predictive ability.




```
Logistic Regression Accuracy: 0.513, AUC: 0.507
Random Forest Accuracy: 0.493, AUC: 0.506
```

Interpretation:
The feature importance plot from the Random Forest model shows that 'self_employed_yes' and 'gender_female' are the most influential variables in predicting treatment-seeking behavior, followed by age groups 46+, 36–45, and 26–35.
While these features do contribute to the model's decisions, their influence is relatively modest, and the overall model performance remains weak. This means that although some demographic factors show minor predictive value, they are not strong enough to build a reliable classifier on their own.

2.4.1 Conclusion :

Based on the results, we cannot accurately classify whether an individual will seek treatment using the given survey features. Both Logistic Regression and Random Forest models performed poorly, with accuracies close to random guessing (0.51 and 0.49) and AUC values of 0.51, indicating no meaningful predictive power.
Despite identifying some influential features like gender and self-employment status, the models failed to reliably distinguish between those who seek treatment and those who do not.

7. Discussion & Insights

7.1 Key Findings

- **Gender is significant**: Females were statistically more likely to seek treatment ($p < 0.05$), while self-employment and age showed weaker effects.
- **Work setting impacts support**: On-site and non-tech employees had significantly better access to mental-health benefits than remote or tech-only workers ($p < 0.01$).
- **Company size matters**: Mid-sized companies (26–100 employees) had the highest treatment rates (~60%) due to a balance of resources and personal support. Small firms

lacked resources; large firms lacked human connection.

7.2 Practical Recommendations

- **Support female employees**: Tailor mental-health resources with gender-sensitive approaches.
- **Expand remote and tech-worker programs**: Ensure equal access to support regardless of work location or company type.
- **Adopt mid-size firm practices**: Balance structured programs with personal support to improve outcomes.
- **Enhance future data collection**: Include more detailed workplace and mental-health variables to enable better prediction and targeting.

8. Conclusion

Mid-sized companies (26-100 employees) achieve the highest mental health treatment rates (~60%) by optimally balancing institutional resources and human support, while demographic analysis reveals gender is the only significant predictor (females 1.8x more likely to seek treatment), workplace factors show remote workers experience 15-20% lower benefit access ($z=3.51$, $p<0.001$) and tech employees receive 12-18% less support than non-tech peers ($z=3.06$, $p=0.002$), with large enterprises (500-1000 employees) paradoxically showing low treatment rates (~45%) despite strong structural support due to eroding human connections—correlation analysis confirming benefits access drives treatment most strongly ($r=+0.92$) while coworker support shows negative correlation ($r=-0.85$)—though predictive modeling failed (AUC=0.51), highlighting that effective mental health strategies must bridge the resource-relationship gap through tailored solutions for different organizational contexts.

8.1 Summary

- Data science helped reveal impactful insights about mental health in the tech field

- Models were able to predict treatment-seeking behavior with moderate accuracy

9. References

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